

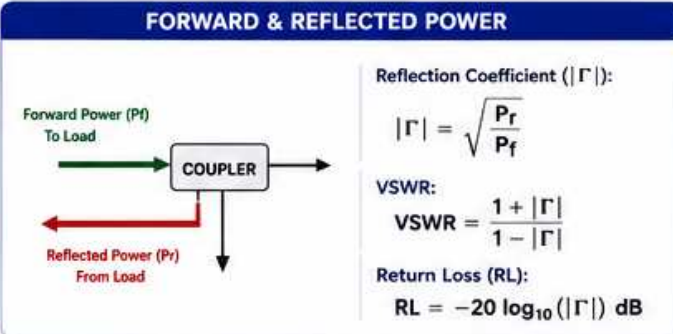
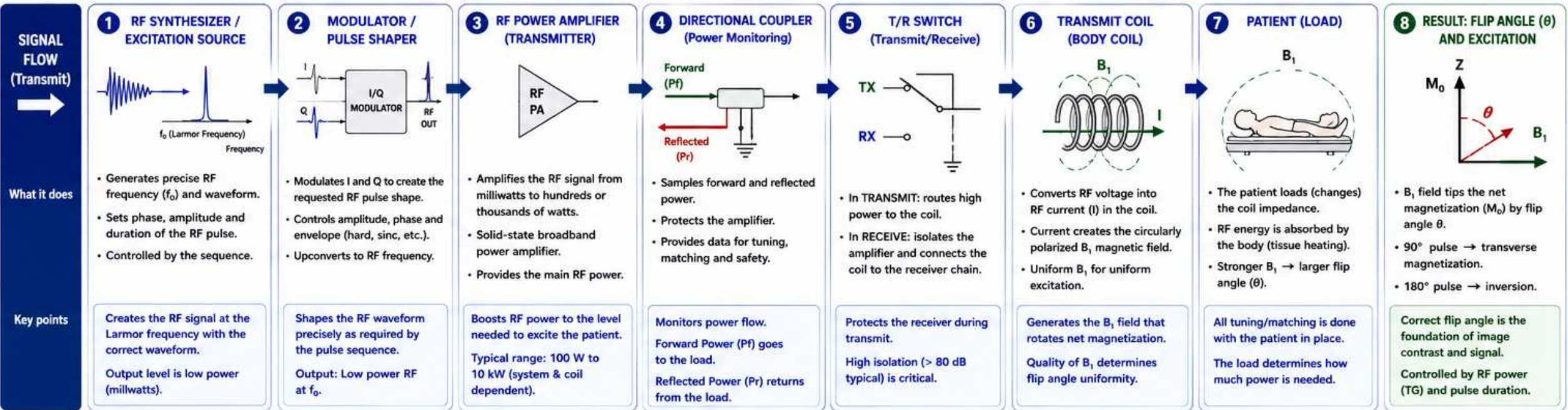


GE MRI

# MRI RF TRANSMIT CHAIN – FROM RF WAVEFORM TO B<sub>1</sub> EXCITATION

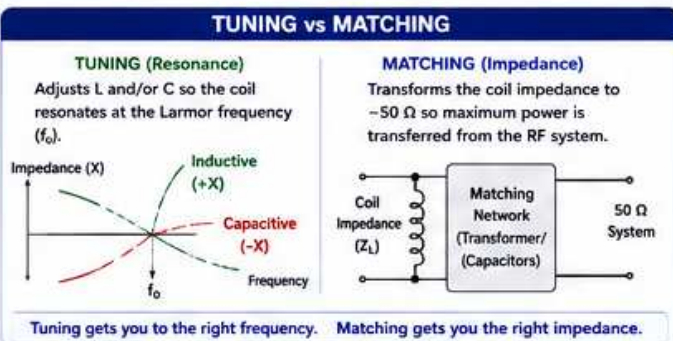
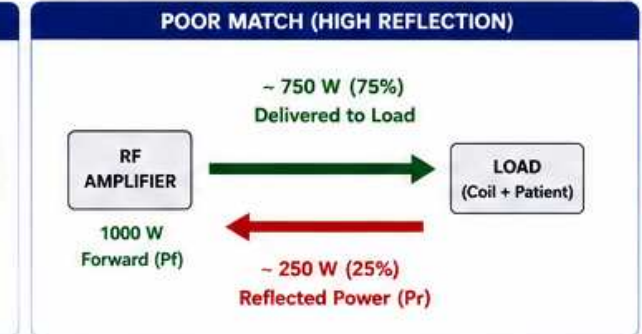
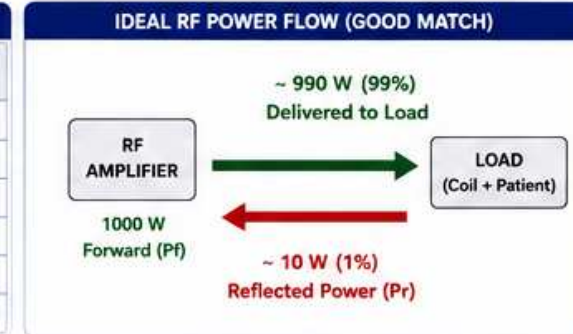
The path of RF energy during transmit: RF waveform → RF power → Coil current → B<sub>1</sub> field → Flip angle

GOAL: Deliver the right amount of RF power at the correct frequency to create the desired B<sub>1</sub> field for accurate flip angles and image contrast.



### MATCH QUALITY GUIDE

| VSWR    | Reflected Power (Approx.) | Return Loss (dB) | Match Quality    |
|---------|---------------------------|------------------|------------------|
| 1.0 : 1 | 0 %                       | ∞                | Perfect          |
| 1.2 : 1 | ~0.8 %                    | ≥ 26 dB          | Excellent        |
| 1.5 : 1 | ~4 %                      | ≥ 14 dB          | Good             |
| 2.0 : 1 | ~11 %                     | ≥ 9.5 dB         | Fair             |
| 3.0 : 1 | ~25 %                     | ≥ 6 dB           | Poor             |
| ∞ : 1   | 100 %                     | 0 dB             | Total Reflection |



### MEASUREMENT POINTS (ENGINEERING PAGES)

- Forward Power (Pf): Measured at coupler forward port.
- Reflected Power (Pr): Measured at coupler reflected port.
- VSWR / Return Loss: Computed from Pf and Pr.
- TG (Transmit Gain / AX): Set by the system during prescan based on patient loading and desired flip angle.
- Coil Impedance: Viewed on service/engineering pages after tuning (at f<sub>0</sub>).

Example (GE Engineering Page)

|                                      |                |
|--------------------------------------|----------------|
| Forward Power (Pf):                  | 1000 W         |
| Reflected Power (Pr):                | 10 W           |
| VSWR:                                | 1.22 : 1       |
| Return Loss:                         | 26.0 dB        |
| TG (AX):                             | 23 dB          |
| Coil Impedance (at f <sub>0</sub> ): | 49.8 Ω ∠ -2.1° |

Status: All Parameters Within Limits

### COMMON FAILURE POINTS (TRANSMIT CHAIN)

|                                          |                                       |
|------------------------------------------|---------------------------------------|
| ⚠ Bad or loose coil cable / connector    | ⚠ T/R switch failure (leakage in TX)  |
| ⚠ Open / shorted transmit coil element   | ⚠ Damaged RF amplifier                |
| ⚠ Coil detuning failure (during receive) | ⚠ Directional coupler drift / failure |
| ⚠ Poor tuning (mistuned resonance)       | ⚠ High reflected power trips          |
| ⚠ Poor matching (high VSWR)              | ⚠ Overheating / poor cooling          |

- ### ENGINEER NOTES
- Always tune and match with the patient (or phantom) in place.
  - High reflected power reduces efficiency and can damage the transmitter.
  - Excessive RF power increases SAR (Specific Absorption Rate) in the patient.

### THE BOTTOM LINE

Efficient RF transmit requires: Right Frequency + Proper Tuning + Good Matching → Maximum Power to the Patient → Correct B<sub>1</sub> → Correct Flip Angle

- ### SAFETY REMINDER
- RF power can cause tissue heating.
  - Follow all SAR, duty cycle and service safety procedures.
  - Verify cooling and interlocks are functioning.

Remember: Power in is good. Power reflected is wasted.